

• Given : Observation $y_{M \times 1}$

• Sensing matrix $A_{M \times N}$

• s , such that $x_{N \times 1}$ is s -sparse.

Req : find a solution to $y = Ax$

• $\circ$ $M < N$ & assuming A has full row rank,

$\circ$ there are infinite possible solutions.

• But we know that x is s -sparse, so we use this to pick the correct solution.

• $\circ$ $y = \sum_{i \in S_0} a_i x_i$, where $S_0 = \{i \mid x_i \neq 0\}$,

$\circ$ only a few columns of A will have high value of the inner product $y^T A$.

• N.B : If we know S_0 , we can solve :

• $y = A_{M \times S} x_{S \times 1}$ & get a unique solution.

• OMP first estimates S_0 , then uses it to get x .

* OMP : Input y, A & s .

• Output : $\hat{x}$ & $\hat{S}_s$.

① Normalize A : $A \leftarrow \begin{bmatrix} \frac{a_1}{\|a_1\|} & \dots & \frac{a_n}{\|a_n\|} \end{bmatrix}$.

② Initialize residual $r=y$ & dictionary $\mathcal{L} = []$ (empty matrix).

③ Initialize $\hat{x} = 0_{n \times 1}$ & $\hat{S}_0 = []$.

④ For j in $(1, 1000, 5)$:

(a) compute the inner product between r & A as $(r^T A)_{1 \times n}$.

(b) Find the index i^* from $1, \dots, n$ that maximizes $(r^T A)$.

(c) Append a_{i^*} to $\mathcal{L}$: $\mathcal{L} \rightarrow [\mathcal{L} \ a_{i^*}]$.

. Note : These steps estimate $\hat{S}_0$ iteratively, i.e.,
 $\hat{S}_0 = [\hat{S}_0 \ i^*]$.

(d) Solve a LS problem to get the values of the non-zero elements of $\hat{x}$:

$$\hat{x}_{\text{small}} = \arg \min_x \|y_{m \times 1} - \mathcal{L}_{m \times j} x_{j \times 1}\|_2^2.$$

$$(e) \hat{x} = \begin{cases} 0, & i \notin \hat{S}_0 \\ \hat{x}_{\text{small}}, & i \in \hat{S}_0. \end{cases}$$

(f) Estimate $\hat{y} = \mathcal{L} \hat{x}$ & $r \leftarrow r - \hat{y}$.